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# Advanced Expense Tracker CLI Application

expenses = {}
expense_id_counter = 1


# Load expenses from file
def load_expenses():
    global expense_id_counter
    try:
        with open("expenses.txt", "r") as file:
            expenses.clear()
            for line in file:
                eid, title, amount, category = line.strip().split("::")
                expenses[int(eid)] = {
                    "title": title,
                    "amount": float(amount),
                    "category": category
                }

            if expenses:
                expense_id_counter = max(expenses.keys()) + 1

    except FileNotFoundError:
        pass


# Save expenses to file
def save_expenses():
    with open("expenses.txt", "w") as file:
        for eid, info in expenses.items():
            file.write(f"{eid}::{info['title']}::{info['amount']}::{info['category']}\n")


# Display expenses
def view_expenses():
    if not expenses:
        print("\nNo expenses recorded.")
        return

    total = 0
    print("\nYour Expenses:")
    for eid, info in expenses.items():
        print(f"[{eid}] {info['title']} | Rs.{info['amount']} | {info['category']}")
        total += info["amount"]

    print(f"\nTotal Spending: Rs.{total}")


# Add expense
def add_expense():
    global expense_id_counter
    title = input("Enter expense title: ").strip()

    try:
        amount = float(input("Enter amount: "))
        category = input("Enter category (Food/Travel/etc): ").strip()

        if title and category:
            expenses[expense_id_counter] = {
                "title": title,
                "amount": amount,
                "category": category
            }
            print(f"Expense added with ID {expense_id_counter}")
            expense_id_counter += 1
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        else:
            print("Fields cannot be empty.")

    except ValueError:
        print("Enter a valid amount.")


# Delete expense
def delete_expense():
    view_expenses()
    if not expenses:
        return

    try:
        eid = int(input("Enter Expense ID to delete: "))
        if eid in expenses:
            print(f"Removed: {expenses[eid]['title']}")
            del expenses[eid]
        else:
            print("Expense ID not found.")
    except ValueError:
        print("Enter a valid number.")


# Category-wise analysis
def category_analysis():
    if not expenses:
        print("\nNo data for analysis.")
        return

    summary = {}
    for info in expenses.values():
        cat = info["category"]
        summary[cat] = summary.get(cat, 0) + info["amount"]

    print("\nSpending by Category:")
    for cat, amt in summary.items():
        print(f"{cat}: Rs.{amt}")


# Menu
def menu():
    print("\n====== EXPENSE TRACKER MENU ======")
    print("1 -> View Expenses")
    print("2 -> Add Expense")
    print("3 -> Delete Expense")
    print("4 -> Category Analysis")
    print("5 -> Save and Exit")


# Main
def main():
    load_expenses()

    while True:
        menu()
        choice = input("Choose option: ").strip()

        if choice == "1":
            view_expenses()
        elif choice == "2":
            add_expense()
        elif choice == "3":
            delete_expense()
        elif choice == "4":
            category_analysis()
        elif choice == "5":
            save_expenses()
            print("Data saved. Exiting program.")
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            print("Data saved. Exiting program.")
            break
        else:
            print("Invalid choice. Try again.")


if __name__ == "__main__":
    main()
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====== EXPENSE TRACKER MENU ======
1 -> View Expenses
2 -> Add Expense
3 -> Delete Expense
4 -> Category Analysis
5 -> Save and Exit
Choose option: 1

No expenses recorded.

====== EXPENSE TRACKER MENU ======
1 -> View Expenses
2 -> Add Expense
3 -> Delete Expense
4 -> Category Analysis
5 -> Save and Exit
Choose option: 2
Enter expense title: Infrastructure
Enter amount: 20,00000
Enter a valid amount.

====== EXPENSE TRACKER MENU ======
1 -> View Expenses
2 -> Add Expense
3 -> Delete Expense
4 -> Category Analysis
5 -> Save and Exit
Choose option: 4

No data for analysis.

====== EXPENSE TRACKER MENU ======
1 -> View Expenses
2 -> Add Expense
3 -> Delete Expense
4 -> Category Analysis
5 -> Save and Exit
Choose option: 4

No data for analysis.

====== EXPENSE TRACKER MENU ======
1 -> View Expenses
2 -> Add Expense
3 -> Delete Expense
4 -> Category Analysis
5 -> Save and Exit
Choose option: 2
Enter expense title: Paintings & Artistry
Enter amount: 10,00000
Enter a valid amount.

====== EXPENSE TRACKER MENU ======
1 -> View Expenses
2 -> Add Expense
3 -> Delete Expense
4 -> Category Analysis
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